Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A bulb is rated at $100\ \text{V}, 100\ \text{W}$, it can be treated as a resistor. Find out the inductance of an inductor (called choke coil) that should be connected in series with the bulb to operate the bulb at its rated power with the help of an ac source of $200\ \text{V}$ and $50\ \text{Hz}$

(A) $\frac{\pi}{\sqrt{3}}\text{H}$

(B) 100H

(C) $\frac{\sqrt{2}}{\pi}\text{H}$

(D) $\frac{\sqrt{3}}{\pi}\text{H}$

Q.2

The instantaneous values of alternating current and voltages in a circuit are given as

$i = \frac{1}{\sqrt{2}}\sin(100\pi t)$ ampere

$e = \frac{1}{\sqrt{2}}\sin(100\pi t + \pi/3)$ Volt

The average power in Watts consumed in the circuit is:

(A) $\frac{1}{4}$

(B) $\frac{\sqrt{3}}{4}$

(C) $\frac{1}{2}$

(D) $\frac{1}{8}$

Q.3

A coil has resistance 30 ohm and inductive reactance 20 ohm at $50\ \text{Hz}$ frequency. If an ac source, of 200 volt, $100\ \text{Hz}$, is connected across the coil, the current in the coil will be

(A) $4.0\ \text{A}$

(B) $8.0\ \text{A}$

(C) $\frac{20}{\sqrt{13}}\ \text{A}$

(D) $2.0\ \text{A}$

Q.4

In an electrical circuit R, L, C and an a.c. voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage the current in the circuit is $\pi/3$. If instead, C is removed from the circuit, the phase difference is again $\pi/3$. The power factor of the circuit is :

@IITJEE_Advanced

(A) $1/2$

(B) $1/\sqrt{2}$

(C) 1

(D) $\sqrt{3}/2$

Q.5

A $100\mu\mathrm{F}$ capacitor in series with a 40Ω resistance is connected to a $110\ \mathrm{V}, 60\ \mathrm{Hz}$ supply. What is the maximum current in the circuit?

(A) $3.24\ \mathrm{A}$

(B) $4.25\ \mathrm{A}$

(C) $2.25\ \mathrm{A}$

(D) $5.20\ \mathrm{A}$

Q.6

An AC generator of $220\ \mathrm{V}$ having internal resistance $\mathrm{r} = 10\Omega$ and external resistance $\mathrm{R} = 100\Omega$. What is the power developed in the external circuit?

(A) $484\ \mathrm{W}$

(B) $400\ \mathrm{W}$

(C) $441\ \mathrm{W}$

(D) $369\ \mathrm{W}$

Q.7

In a series LCR circuit $R = 200\Omega$ and the voltage and the frequency of the main supply is $220\ \mathrm{V}$ and $50\ \mathrm{Hz}$ respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30°. On taking out the inductor from the circuit the current leads the voltage by 30°. Then what will be the power dissipated (in W) in the LCR circuit?

(A) 242 W

(B) 240 W

(C) 239 W

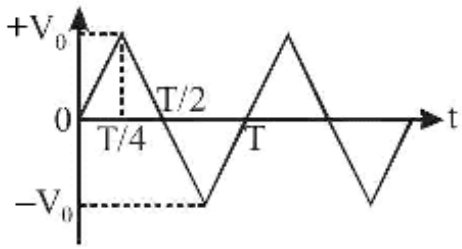
(D) 243 W

Q.8

The voltage time (V-t) graph for triangular wave having peak value V_0 is as shown in figure. The rms value of V

@IITJEE_Advanced

in time interval from $t = 0$ to $T/4$ is $\frac{V_0}{\sqrt{x}}$ then find the value of x.



(A) 3

(B) 2

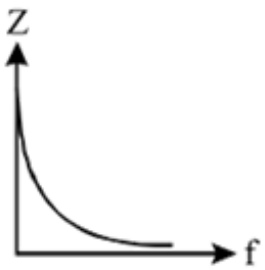
(C) 1

(D) 4

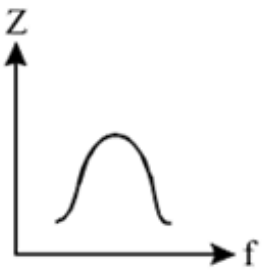
Q.9

Which one of the following curves represents the variation of impedance (Z) with frequency f in series LCR circuit?

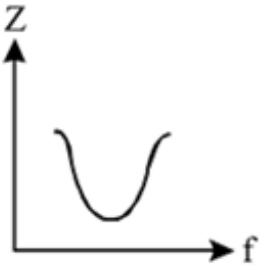
(A)



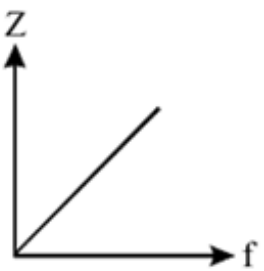
(B)



(C)



(D)



@IITJEE_Advanced

Q.10

Current in an ac circuit is given by $\mathrm{i} = 3\sin\omega\mathrm{t} + 4\cos\omega t$ then

(A) rms value of current is $5\ \mathrm{A}$

(B) mean value of this current in one half period will be $6/\pi$

(C) if voltage applied is $\mathrm{V} = \mathrm{V_m}\sin\omega\mathrm{t}$ then the circuit must be containing resistance and capacitance.

(D) if voltage applied is $\mathrm{V} = \mathrm{V_m}\sin\omega\mathrm{t}$, the circuit may contain resistance and inductance.

Q.11

If we increase the driving frequency in a circuit with a purely capacitive load, then

(A) amplitude V_C increases

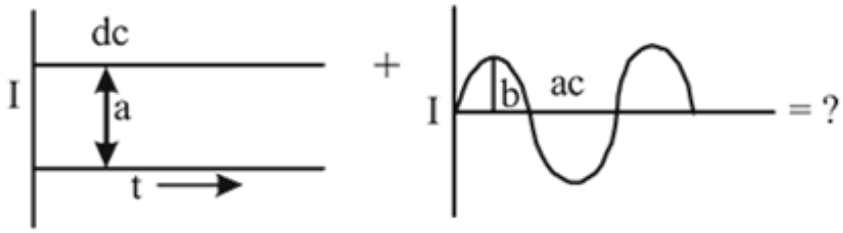
(B) amplitude V_C decreases

(C) amplitude i_C increases

(D) amplitude i_C decreases

Q.12

If a direct current of value ampere is superimposed on an alternating current $\mathrm{I} = \mathrm{b}\sin\omega t$ flowing through a wire, what is the effective value of the resulting current in the circuit?



(A) $\left[\mathrm{a}^2 - \frac{1}{2}\,\mathrm{b}^2\right]^{1/2}\ \mathrm{A}$

(B) $\left[a^2 + b^2\right]^{1/2}\ \mathrm{A}$

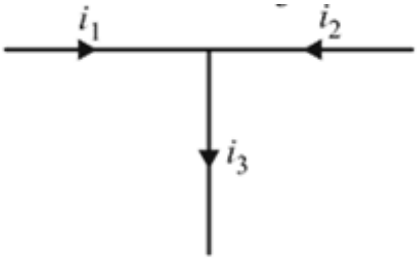
(C) $\left[\frac{a^2}{2} + b^2\right]^{1/2}\ \mathrm{A}$

(D) $\left[a^2 + \frac{b^2}{2}\right]^{1/2}\ \mathrm{A}$

Q.13

@IITJEE_Advanced

If $i_1 = 3\sin\omega t$ and $i_2 = 4\cos\omega t$, then i_3 is



(A) $5\sin(\omega t + 53^\circ)$

(B) $5\sin(\omega t + 37^\circ)$

(C) $5\sin(\omega t + 45^\circ)$

(D) $5\cos(\omega t + 53^\circ)$

Q.14

A sinusoidal voltage $\mathrm{V(t) = 100\sin(500t)}$ is applied across a pure inductance of $\mathrm{L = 0.02H}$. The current through the coil is:

(A) $\mathrm{10\cos(500t)}$

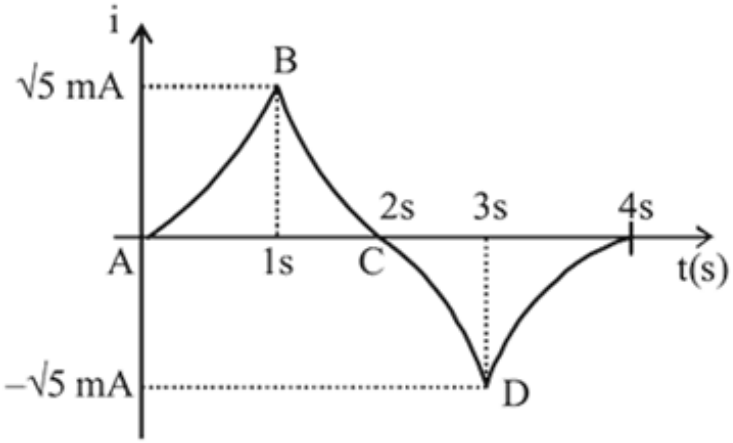
(B) $\mathrm{-10\cos(500t)}$

(C) $\mathrm{10\sin(500t)}$

(D) $\mathrm{-10\sin(500t)}$

Q.15

Figure shows one cycle of an alternating current with the segments $\mathrm{AB, BC, CD, DE}$ being symmetrical and parabolic. The root mean square value of this current over one cycle is $\mathrm{x mA}$, find x.



(A) 1mA

(B) 2mA

(C) 3mA

(D) 4mA

Q.16

@IITJEE_Advanced

Resonance frequency of a circuit is f. If the capacitance is made 4 times the initial value, then the resonance frequency will become:

(A) $\mathrm{f}/2$

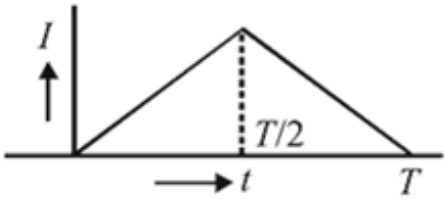
(B) $2\mathrm{f}$

(C) f

(D) $\mathrm{f}/4$

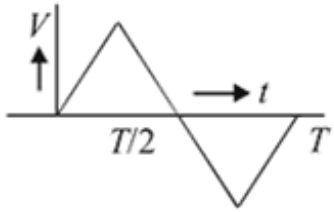
Q.17

The current (I) in the inductance is varying with time according to the plot shown in figure.

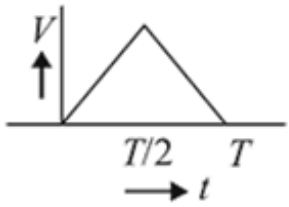


Which one of the following is the correct variation of voltage with time in the coil?

(A)



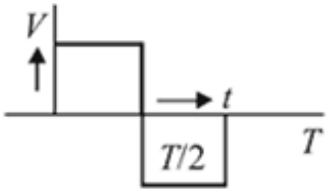
(B)



(C)



(D)



Q.18

The voltage of an ac supply varies with time (t) as $V = 120 \sin 100\pi t \cos 100\pi t$. The maximum voltage and frequency respectively are

(A) 120 volts, 100 Hz

@IITJEE_Advanced

(B) $\frac{120}{\sqrt{2}}$ volts, $100\ \mathrm{Hz}$

(C) 60 volts, $200\ \mathrm{Hz}$

(D) 60 volts, $100\ \mathrm{Hz}$

Q.19

The equation of AC voltage is $E = 220\sin(\omega t - \pi/6)$ and the A.C. current is $I = 10\sin(\omega t + \pi/6)$. The average power dissipated is

(A) $150\ \mathrm{W}$

(B) $550\ \mathrm{W}$

(C) $250\ \mathrm{W}$

(D) $50\ \mathrm{W}$

Q.20

In an ac circuit an alternating voltage $\mathrm{e} = 200\sqrt{2}\sin 100\mathrm{t}$ volts is connected to a capacitor of capacitance $1\mu\mathrm{F}$. The r.m.s. value of the current in the circuit is

(A) $10\ \mathrm{mA}$

(B) $100\ \mathrm{mA}$

(C) $200\ \mathrm{mA}$

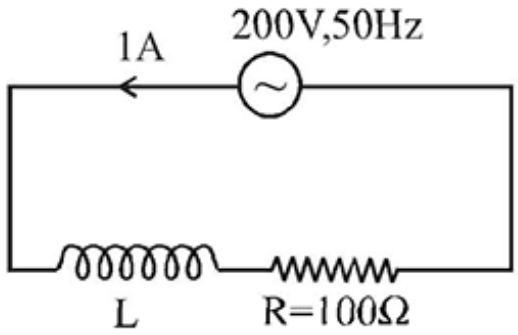
(D) $20\ \mathrm{mA}$

Answers & Solutions

Q.1 Answer:

$\frac{\sqrt{3}}{\pi}\mathrm{H}$

Solution:



From the rating of the bulb, the resistance of the bulb can be calculated.

$\mathrm{R} = \frac{\mathrm{V}_{\mathrm{rms}}^2}{\mathrm{P}} = 100\Omega$

For the bulb to be operated at its rated value the rms **current through it should be** $1\ \mathrm{A}$

$\text{Also, } \mathrm{I}_{\mathrm{rms}} = \frac{\mathrm{V}_{\mathrm{rms}}}{\mathrm{Z}}$

@IITJEE_Advanced

$\therefore$

$$1 = \frac{200}{\sqrt{100^2 + (2\pi 50.\mathrm{L})^2}}$$

$$\mathrm{L} = \frac{\sqrt{3}}{\pi}\mathrm{H}$$

Q.2 Answer:

$\frac{1}{8}$

Solution:

The average power in the circuit where $\cos\phi =$ power factory

$\langle P >= V_{\mathrm{rms}} \times I_{\mathrm{rms}} \cos\phi$

$\phi = \pi/3 =$ phase difference $= \frac{180}{3} = 60$

$V_{\mathrm{rms}} = \frac{\frac{1}{\sqrt{2}}}{\sqrt{2}} = \frac{1}{2}\mathrm{volt}$

$I_{\mathrm{rms}} = \frac{1}{\sqrt{2}} = \left(\frac{1}{2}\right) A$

$\cos\phi = \cos\frac{\pi}{3} = \frac{1}{2}$

$< P >= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}\ \mathrm{W}$

Q.3 Answer:

4.0 A

Solution:

If $\omega = 50 \times 2\pi$ then $\omega\mathrm{L} = 20\Omega$

If $\omega' = 100 \times 2\pi$ then $\omega'\mathrm{L} = 40\Omega$

Current flowing in the coil is

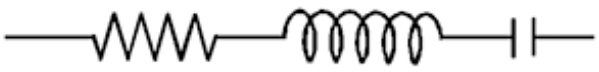
$$I = \frac{200}{Z} = \frac{200}{\sqrt{R^2+(\omega' L)^2}} = \frac{200}{\sqrt{(30)^2+(40)^2}}$$

$$I = 4A.$$

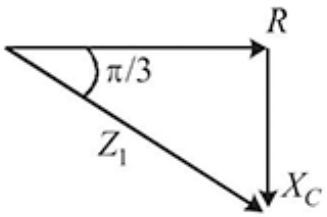
Q.4 Answer:

1

Solution:



when L is removed from the circuit

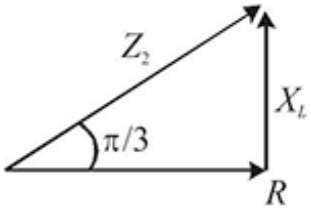


@IITJEE_Advanced

$\frac{X_C}{R} = \tan\frac{\pi}{3}$

$X_C = R\tan\frac{\pi}{3}$ ...(1)

when C is remove from the circuit



$\frac{X_L}{R} = \tan\frac{\pi}{3}$

$X_C = R\tan\frac{\pi}{3}$ ...(2)

net impedence $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$

power factor $\cos\phi = \frac{R}{Z} = 1$

Q.5 Answer:

3.24 A

Solution:

Here, $\mathrm{C} = 100\mu\mathrm{F} = 100 \times 10^{-6}$ F, $\mathrm{R} = 40\Omega$

$\mathrm{V_{ms}} = 110$ V, $\mathrm{f} = 60$ Hz

Peak voltage, $\mathrm{V_0} = \sqrt{2}.\mathrm{V_{rms}} = 100\sqrt{2} = 155.54$ V

Circuit impedance, $\mathrm{Z} = \sqrt{\mathrm{R}^2 + \frac{1}{\omega^2\mathrm{C}^2}}$

$= \sqrt{40^2 + \frac{1}{(2\times\pi\times60\times100\times10^{-6})^2}}$

$= \sqrt{1600 + 703.60} = \sqrt{2303.60} = 48\Omega$

hence, maximum current in coil,

$\mathrm{I_0} = \frac{\mathrm{V_0}}{\mathrm{Z}} = \frac{155.54}{48} = 3.24$ A

Q.6 Answer:

400 W

Solution:

$V = 220$ V; $r = 10\Omega$

$R' = 10 + 100\Omega = 110\Omega$

$I = \frac{V}{R'} = \frac{220}{110} = 2$ A

$P = I^2R = 4 \times 100 = 400$ W

Q.7 Answer:

242 W

Solution:

When capacitance is taken out, the circuit is LR.

$\therefore \tan\phi = \frac{\omega L}{R}$

@IITJEE_Advanced

$\Rightarrow \omega L = R\tan\phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$

Again, when inductor is taken out, the circuit is CR. $\therefore \quad \tan\phi = \frac{1}{\omega CR}$

$\Rightarrow \frac{1}{\omega c} = R\tan\phi = 200 \times \frac{1}{\sqrt{3}} = \frac{200}{\sqrt{3}}$

Now, $Z = \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}$

$= \sqrt{(200)^2 + \left(\frac{200}{\sqrt{3}} - \frac{200}{\sqrt{3}}\right)^2} = 200\Omega$

Power dissipated $= V_{rms} I_{rms} \cos\phi$

$= \mathrm{V_{rms}} \cdot \frac{\mathrm{V_{ms}}}{\mathrm{Z}} \cdot \frac{R}{\mathrm{Z}} \left(\because \cos\phi = \frac{R}{Z}\right)$

$= \frac{V_{\mathrm{rms}}^2 R}{\mathrm{Z}^2} = \frac{(220)^2 \times 200}{(200)^2}$

$= \frac{220 \times 220}{200} = 242\ \mathrm{W}$

Q.8 Answer:

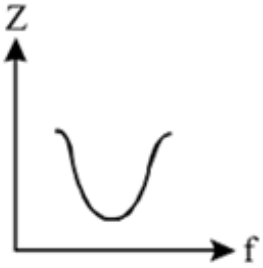
3

Solution:

$\mathrm{V} = \frac{\mathrm{V_0}}{\mathrm{T}/4}\mathrm{t} \Rightarrow \mathrm{V} = \frac{4\,\mathrm{V_0}}{\mathrm{T}}\mathrm{t}$

$\Rightarrow \mathrm{V_{rms}} = \sqrt{< \mathrm{V}^2 >} = \frac{4\,\mathrm{V_0}}{\mathrm{T}}\sqrt{< \mathrm{t}^2 >} = \frac{4\,\mathrm{V_0}}{\mathrm{T}} \left\{\frac{\int_0^{\mathrm{T}/4} \mathrm{t}^2 \mathrm{dt}}{\int_0^{\mathrm{T}/4} \mathrm{dt}}\right\}^{1/2} = \frac{\mathrm{V_0}}{\sqrt{3}}$

Q.9 Answer:



Solution:

Impedance at resonant frequency is minimum in series LCR circuit.

So, $Z = \sqrt{R^2 + \left(2\pi\mathrm{fL} - \frac{1}{2\pi\mathrm{fC}}\right)^2}$

When frequency is increased or decreased. Z increases.

Q.10 Answer:

if voltage applied is $\mathrm{V} = \mathrm{V_m} \sin\omega\mathrm{t}$ then the circuit must be containing resistance and capacitance.

Solution:

@IITJEE_Advanced

$$I = 3\sin\omega t + 4\cos\omega t$$
$$= 5\left(\frac{3}{5}\sin(\omega t) + \frac{4}{5}\cos(\omega t)\right)$$
$$I = 5\sin(\omega t + \alpha)\alpha = -1\left(\frac{4}{5}\right)$$
$$\gamma\mu\int = \frac{5}{\sqrt{2}}$$

Mean value of current $= \frac{1}{\pi}\int_0^{\pi} 5\sin\theta\, d\theta = \frac{10}{\pi}$**, current lags voltage so contain R and C only.**

Q.11 Answer:

amplitude i_C increases

Solution:

As $X_C = \frac{1}{\omega C}$**, so with the increase in frequency,** X_{C} **decreases and so** i_{C} **increases.**

Q.12 Answer:

$\left[a^2 + \frac{b^2}{2}\right]^{1/2}$ A

Solution:

As current at any instant in the circuit will be

$\mathrm{I} = \mathrm{I_{dc}} + \mathrm{I_{ac}} = \mathrm{a} + \mathrm{b}\sin\omega\mathrm{t}$

So, $\mathrm{I_{eff}} = \left[\frac{\int_0^{\mathrm{T}} \mathrm{I}^2\mathrm{dt}}{\int_0^{\mathrm{T}} \mathrm{dt}}\right]^{1/2} = \left[\frac{1}{\mathrm{T}}\int_0^{\mathrm{T}}(\mathrm{a} + \mathrm{b}\sin\omega\mathrm{t})^2\mathrm{dt}\right]^{1/2}$

i.e. $I_{\mathrm{eff}} = \left[\frac{1}{T}\int_0^T \left(a^2 + 2ab\sin\omega t + b^2\sin^2\omega t\right) dt\right]^{1/2}$

but as $\frac{1}{T}\int_0^T \sin\omega\,\mathrm{tdt} = 0$ **and** $\frac{1}{T}\int_0^T \sin^2\omega t dt = \frac{1}{2}$

So, $I_{\mathrm{eff}} = \left[a^2 + \frac{1}{2}b^2\right]^{1/2}$

Q.13 Answer:

$5\sin(\omega t + 53^\circ)$

Solution:

$i_3 = i_1 + i_2 = 3\sin\omega t + 4\cos\omega t$
$= 5\sin(\omega t + \phi)$, **where** $\tan\phi = \frac{4}{3}$ **or** $\phi = 53^\circ$

Q.14 Answer:

$-10\cos(500\mathrm{t})$

Solution:

In a pure inductive circuit current always lags behind the emf by $\frac{\pi}{2}$ **If** $v(t) = v_0\sin\omega t$ **then** $\mathrm{I} = \mathrm{I_0}\sin\left(\omega t - \frac{\pi}{2}\right)$

Now, given $\mathrm{v(t)} = 100\sin(500\mathrm{t})$

@IITJEE_Advanced

and $I_0 = \frac{E_0}{\omega L} = \frac{100}{500 \times 0.02} [\because \mathrm{L} = 0.02\mathrm{H}]$

$I_0 = 10 \sin\left(500t - \frac{\pi}{2}\right) = -10\cos(500t)$

Q.15 Answer:

1mA

Solution:

RMS value over one cycle = RMS value over AB.

$0 \leq t \leq 15$

$i(t) = \sqrt{5}t^2 c$

$\mathrm{i_{rms}} = \sqrt{< \mathrm{i}^2 >} = \sqrt{\frac{5\int_0^1 t^4 dt}{\int_0^1 dt}}$ **= 1mA**

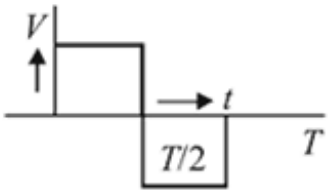
Q.16 Answer:

$\mathrm{f}/2$

Solution:

$\mathrm{f} = \frac{1}{2\pi\sqrt{LC}}$ **i.e.** $f \propto \frac{1}{\sqrt{C}} \rightarrow \frac{1}{\sqrt{4}} = \frac{1}{2}$ **time**

Q.17 Answer:



Solution:

$V = -L\frac{di}{dt}$ **Here** $\frac{di}{dt}$ **is + ve for** $\frac{T}{2}$ **time and** $\frac{di}{dt}$ **is** $-$ **ve for next** $\frac{T}{2}$ **time**

Q.18 Answer:

60 volts, 100 Hz

Solution:

Given, $V = 120\sin(100\pi t)\cos(100\pi t)$

$= 60[2\sin(100\pi t) \cdot \cos(100\pi t)] = 60\sin 200\pi t$

Thus, $V_0 = 60\ \mathrm{V}, \omega = 200\pi$ **or** $f = \frac{200\pi}{2\pi} = 100\ \mathrm{Hz}$.

Q.19 Answer:

550 W

Solution:

We know that, $Z = \frac{E_0}{I_0}$ **Given,** $E_0 = 220$ **and** $I_0 = 10$

so $Z = \frac{220}{10} = 22\ \mathrm{ohm}\ \phi = \left[\frac{\pi}{6} - \left(-\frac{\pi}{6}\right)\right] = \frac{\pi}{3}$

$p_a = \frac{E_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos\phi = \frac{220}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \times \cos\frac{\pi}{3}$

@IITJEE_Advanced

pa $= (220 \times 10 \times 1) / (2 \times 2)$

pa= 550 W

Q.20 Answer:

20 mA

Solution:

$V_{rms} = \frac{200\sqrt{2}}{\sqrt{2}} = 200$ V

$I_{rms} = \frac{V_{rms}}{X_C} = \frac{\frac{200}{1}}{100 \times 10^{-6}} = 2 \times 10^{-2} = 20$ mA

@IITJEE_Advanced